



Better lives through livestock

Pathogen Genomic Identification & Surveillance (Rift Valley fever Virus)

*Author name: **Gilbert Kibet***

*Position: **Research Associate***

*Project: **ILRI Genomics Platform***

*Program: **Health***

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Online documentation: [github/kibet-gilbert/rvf-v-ngs-surveillance](https://github.com/kibet-gilbert/rvf-v-ngs-surveillance)

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Overview

- Genomic surveillance & Why it matters for emerging pathogens
- Recovering viral genomes from clinical samples 
- NGS approaches (Metagenomics vs. Targeted Enrichment)
- Targeted Enrichment through Tiled amplicon sequencing
-  Real-world application: RVFV, Dengue, and Chikungunya
- Lineage assignment and phylogenetic analysis
-  Impact on outbreak response and surveillance

Pathogen Identification and Surveillance Through Genomics

Why Genomic Surveillance?

- Systemic sequencing
- Over time & Place
- Linkage to epidemiological data



Accurate pathogen identification
as compared to
convention
diagnostic tools.



**Track transmission
and evolutionary
changes**
over time and
geography.



**Predict the
potential severity**
of new pathogens
or variants.



**Valuable
information for
drug development**
for vaccines and
therapeutics.

Rift Valley fever Virus

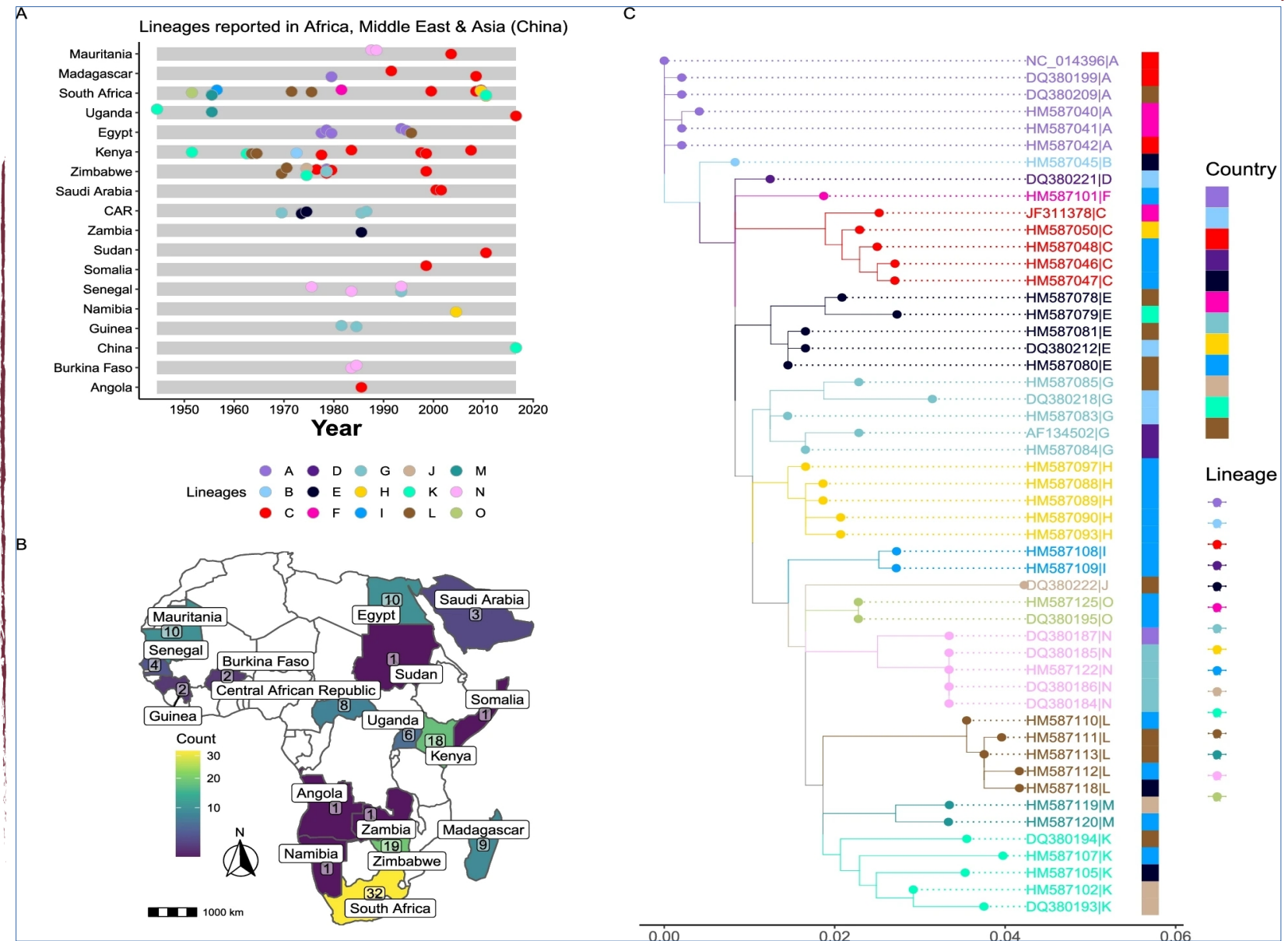
Rift Valley fever Virus (*Phlebovirus riftense*)

- First described 1931 - sheep abortion epidemic in Kenya
- Endemic in sub-Saharan Africa & Arabian Peninsula
- Aedes & Culex mosquitoes
- Documented Epi-zootic & Epidemic events: Egypt (1977), Western Africa (1988) and Arabian Peninsula (2000)
- Humans: 0.5-2.0% fatality ratio
- Livestock: “Abortion Storm” - neonatal mortality (50-100%)



RVFV Distribution

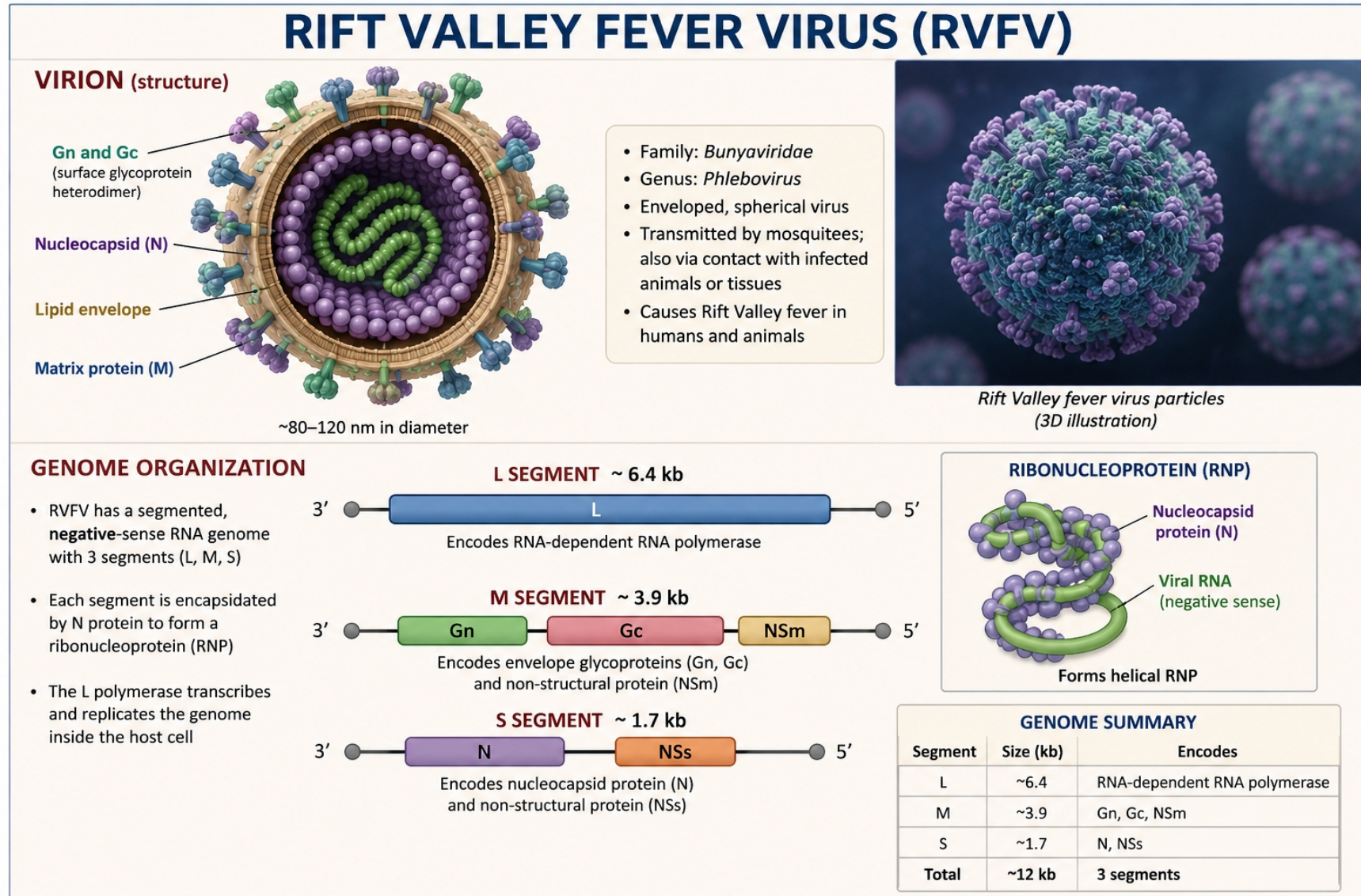
- Lineages: 15 distinct lineages identified in Africa (A through O)
- Lineage C: Most predominant (documented since 1976)
- Lineage H: Second most common in recent outbreaks
- Other lineages: Geographically or temporally restricted



Juma, J., Fonseca, V., Konongoi, S.L. et al. Genomic surveillance of Rift Valley fever virus: from sequencing to lineage assignment. BMC Genomics 23, 520 (2022).

RVFV Genome Organization

- Negative-sense, ssRNA segments
- Low genetic diversity overall.
- 4% nucleotide-level variation (highest: ~5% in M-segment)
- 1% amino acid-level variation
- NO well-defined serotypes

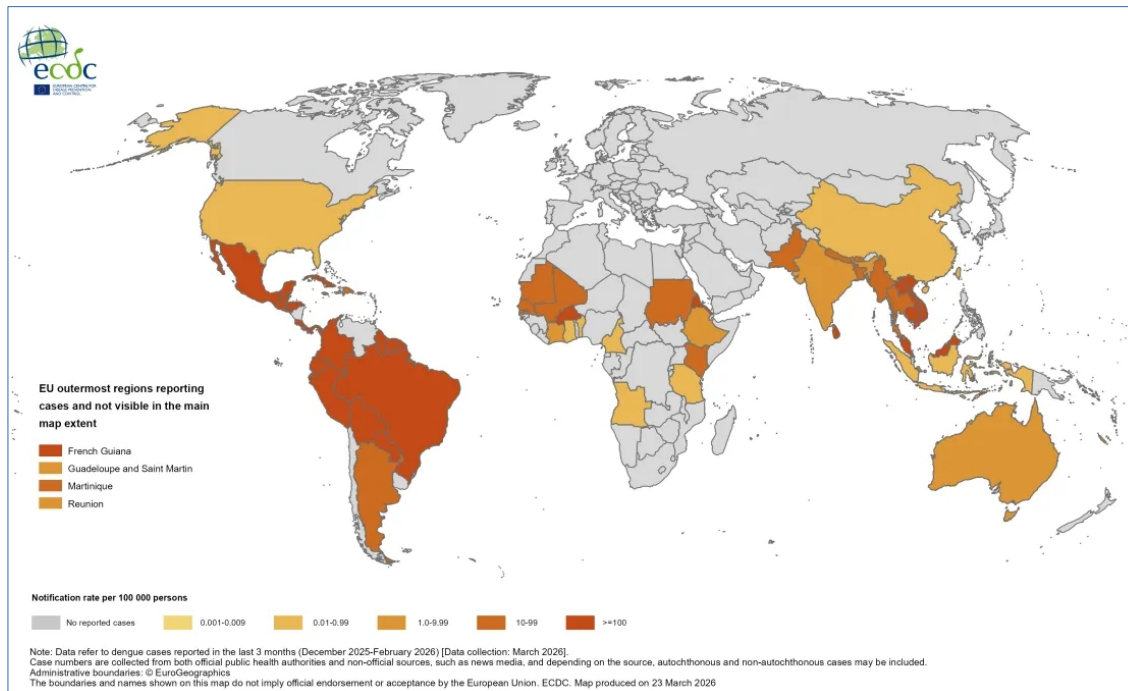


GENOME SUMMARY		
Segment	Size (kb)	Encodes
L	~6.4	RNA-dependent RNA polymerase
M	~3.9	Gn, Gc, NSm
S	~1.7	N, NSs
Total	~12 kb	3 segments

Compared to DENV and CHIKV

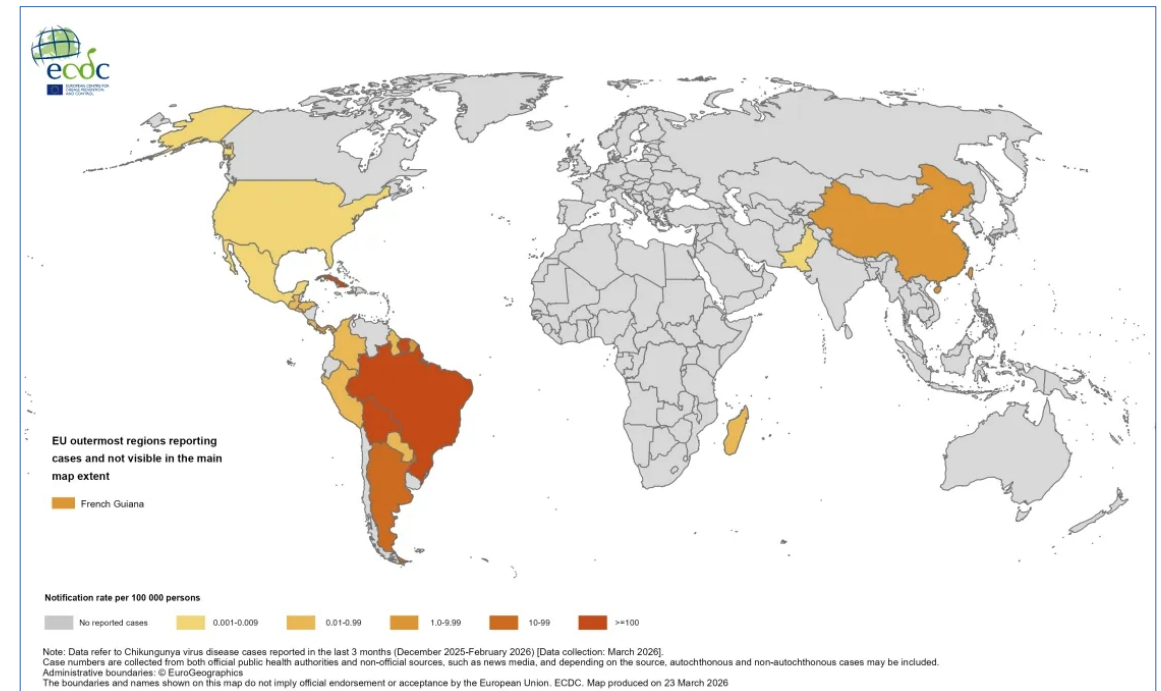
Dengue Virus

- 4 serotypes (DENV-1 to DENV-4)
- +Single-stranded RNA genome (~10.7-11 kb)
- WHO: Top 10 global public health threats



Chikungunya Virus

- Genome: ~11.8 kb +single-stranded RNA
- Characteristic debilitating joint pain (arthralgia)
- Endemic in >110 countries; significant African circulation

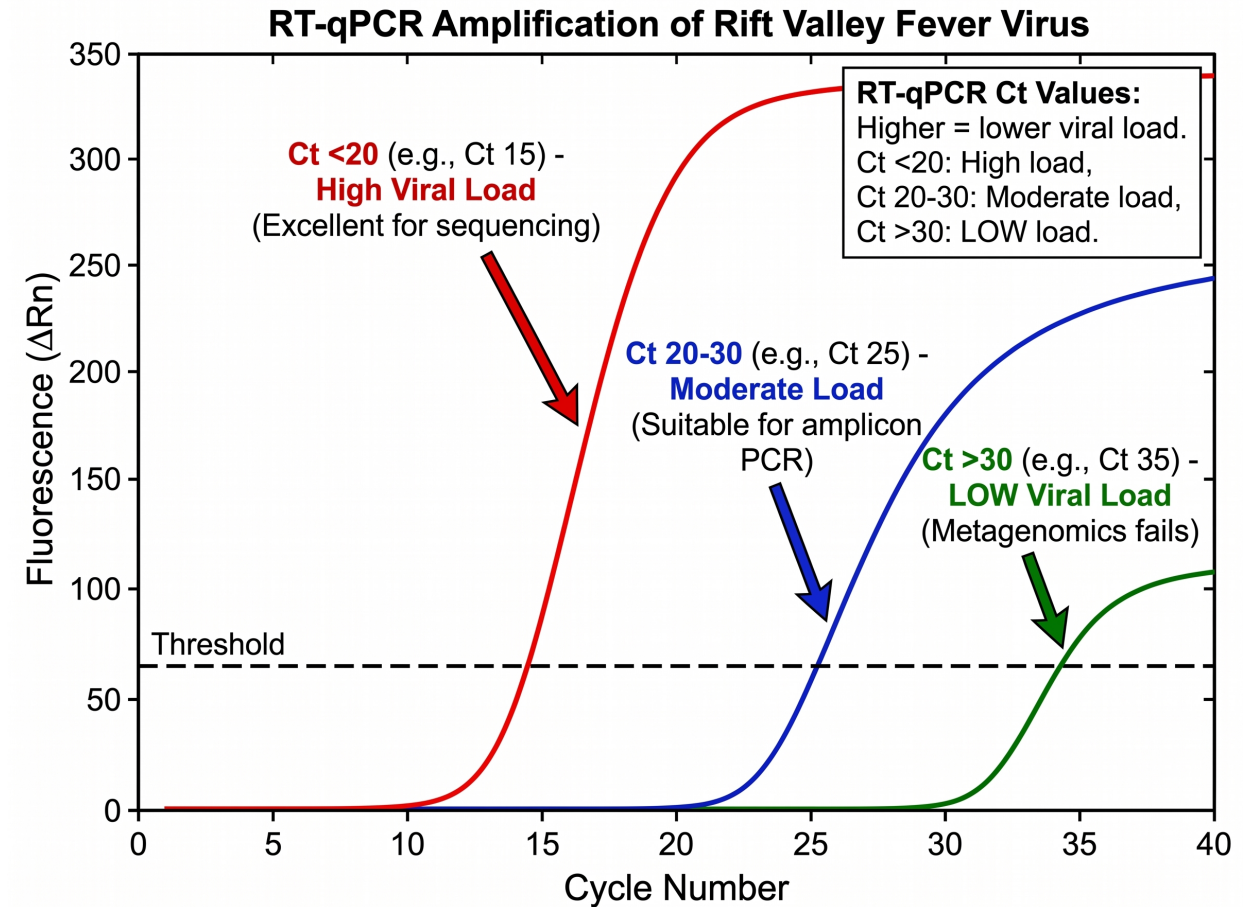


NGS Approaches

NGS Challenges: Low Viral Titres

RT-qPCR Ct values: Higher = lower viral load

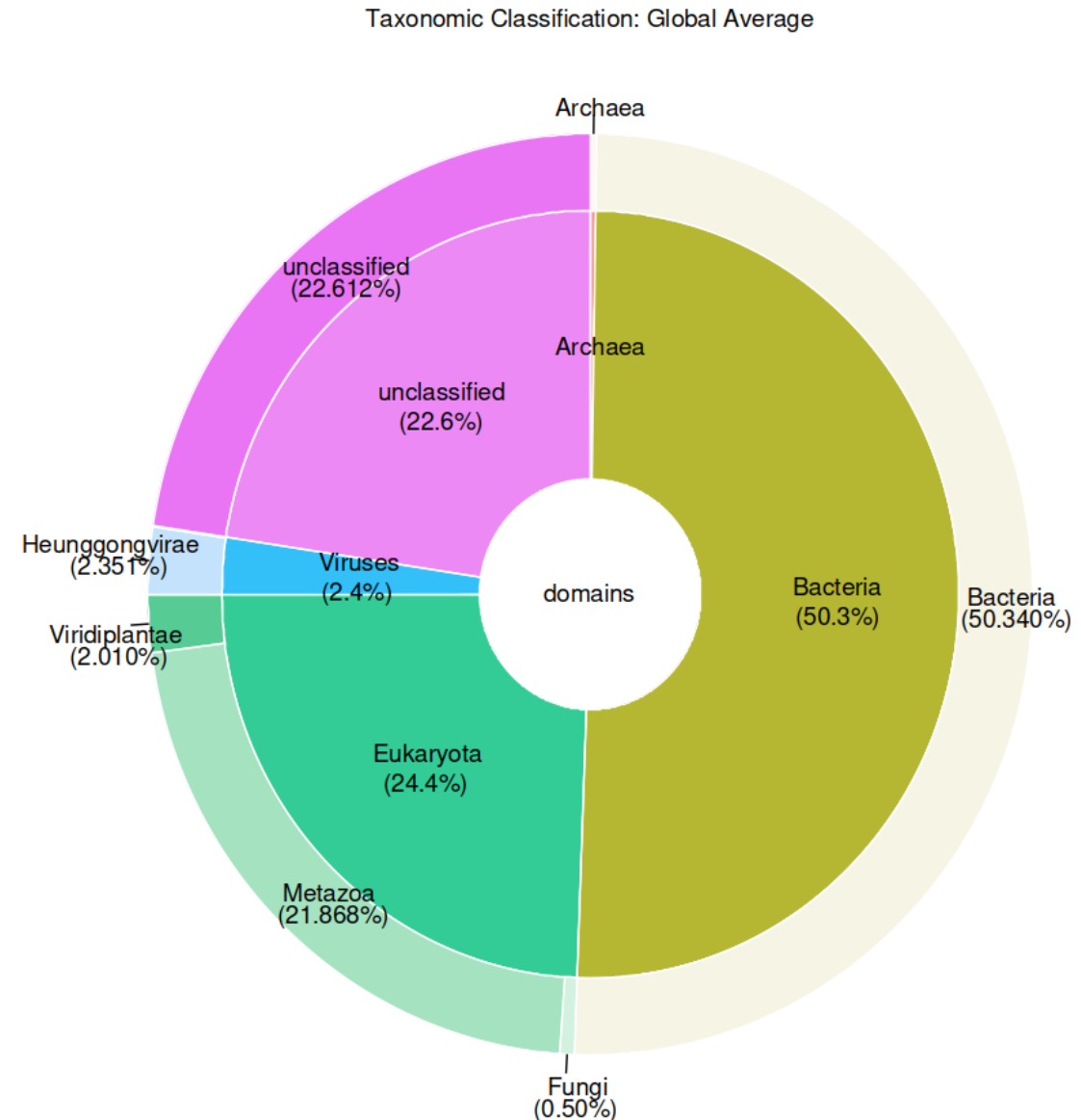
- Ct <20: High viral load (excellent for sequencing)
- Ct 20-30: Moderate load (suitable for amplicon PCR)
- Ct >30: LOW viral load (metagenomics fails)



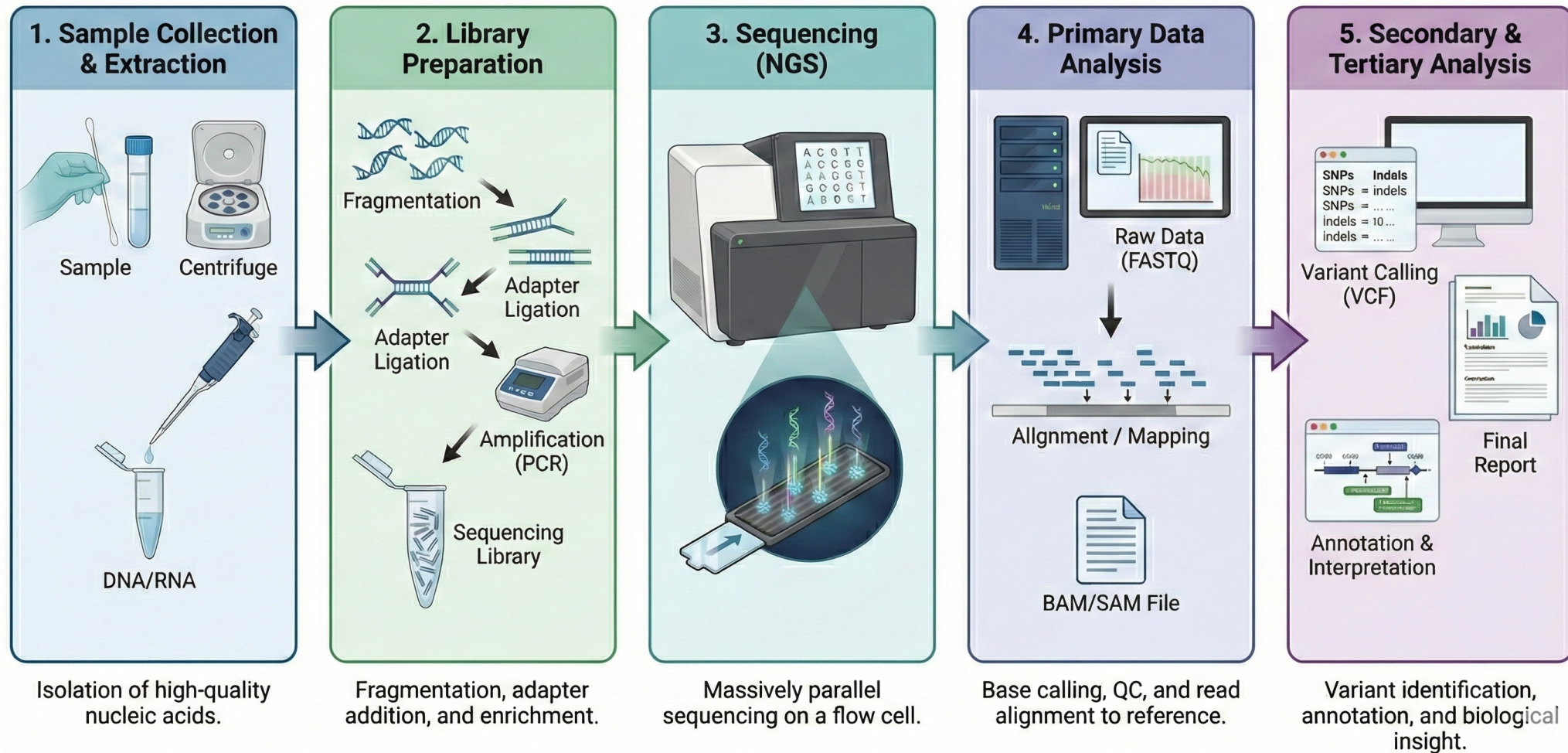
NGS Challenges: RVFV

RVFV Peak viremia in first 5-8 days post-infection

- After 8 days, **viremia drops** dramatically
- Short window for **sequence recovery** from clinical samples
- Host and host-microbiota **contamination**
- **Sample Degradation:** Viral RNA is highly unstable and can degrade quickly if not handled correctly



NGS Overview



NGS Approach 1: Metagenomics

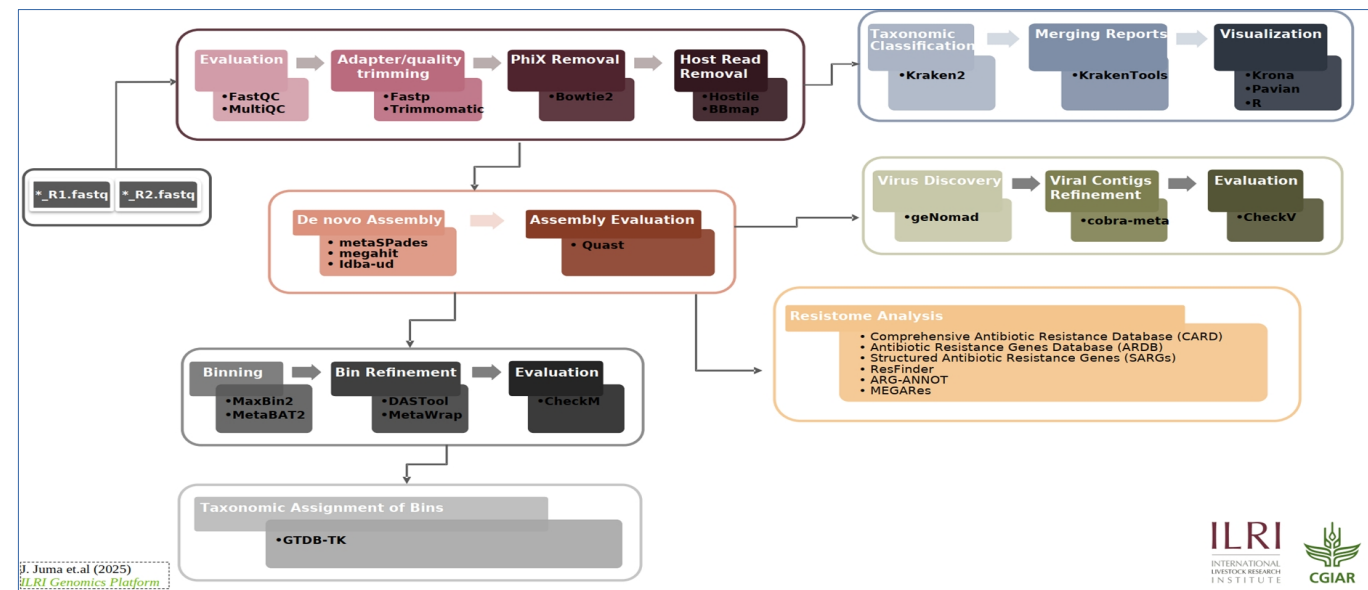
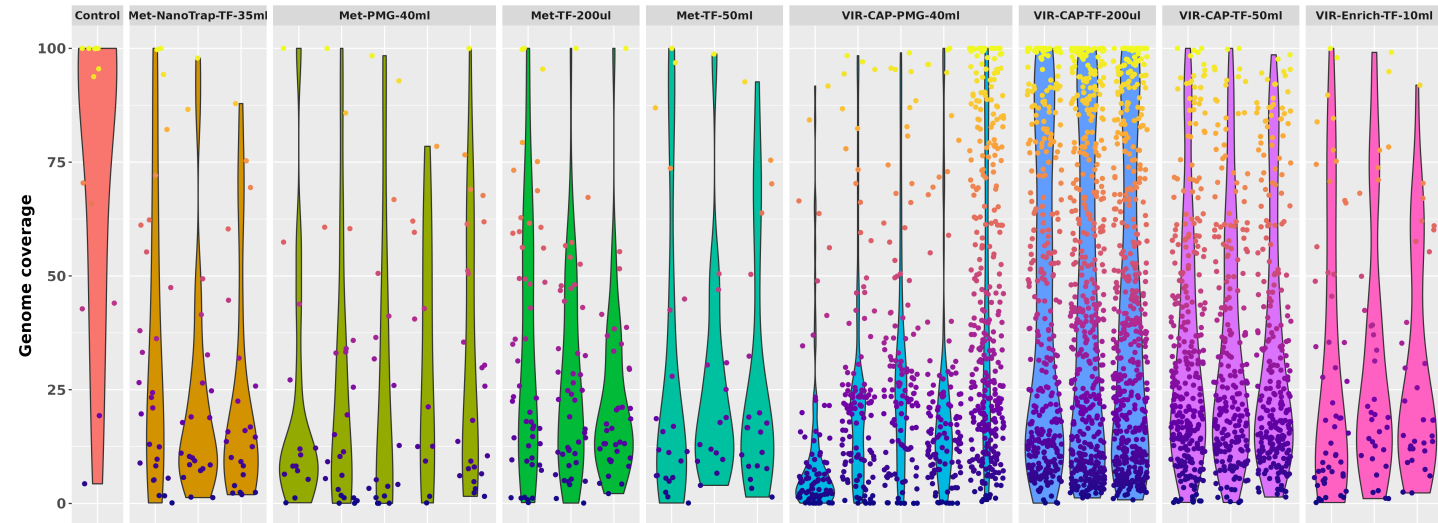
Concept: Sequence Everything in a Sample

Advantages:

- Unbiased - discovers unexpected pathogens
- Useful when etiology UNKNOWN

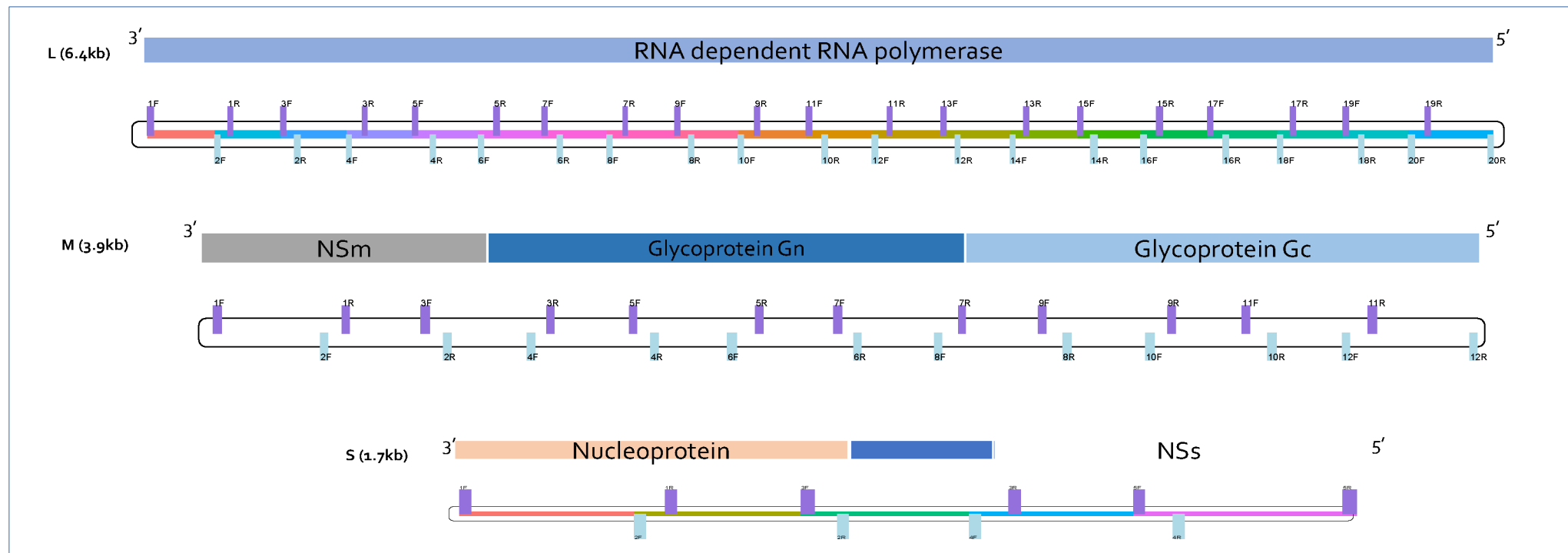
Disadvantages:

- Low sensitivity for low-abundance pathogens
- Expensive (many reads wasted on host)
- Slow analysis (extensive bioinformatic processing)
- Requires high coverage (100M+ reads)



NGS Approach 2: Tiled Amplicons Sequencing

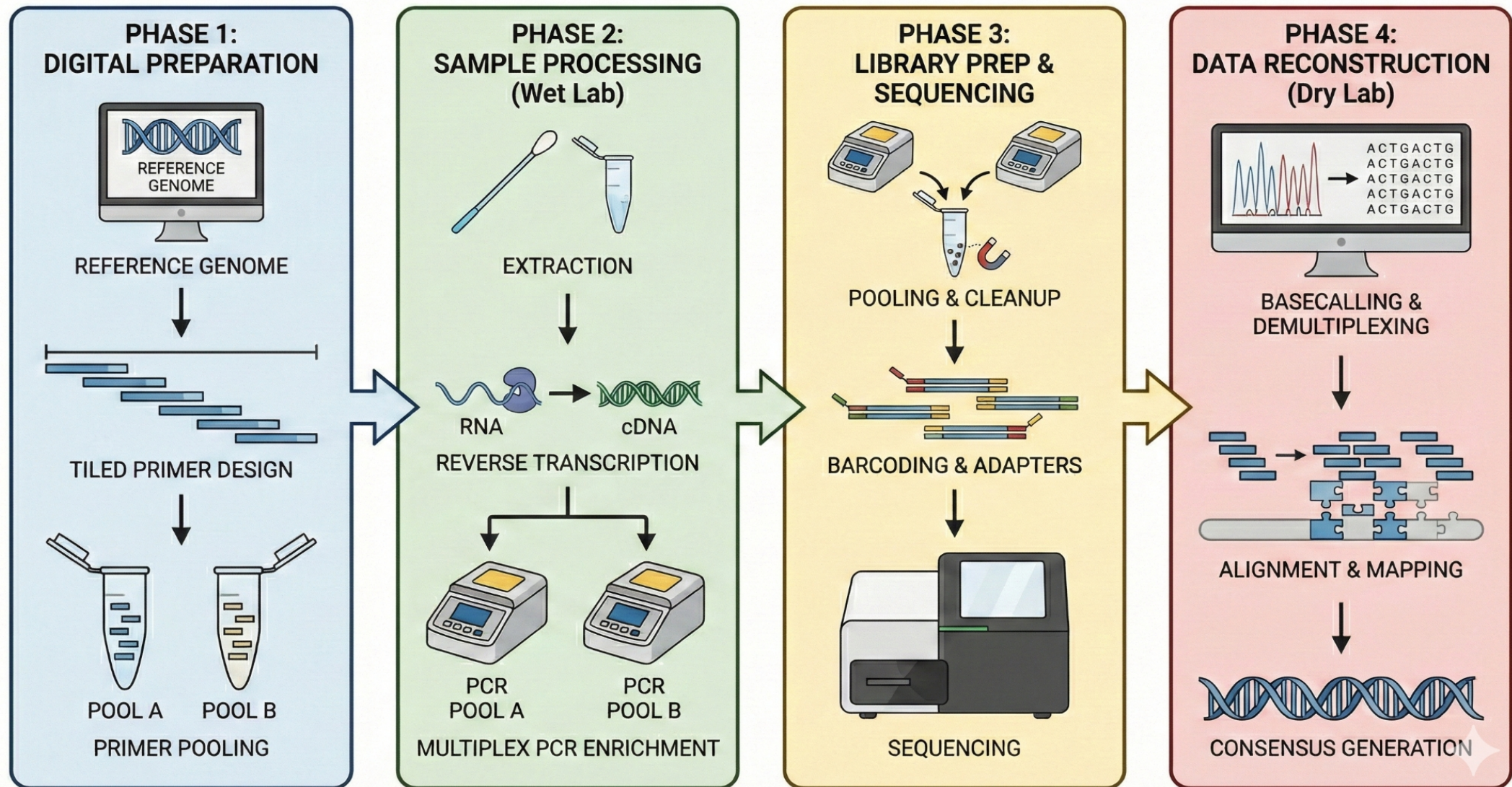
- **Target enrichment** method using multiplex PCR
- Primer pools (“Tiles”) - amplicons overlap - complete coverage of the target genome.



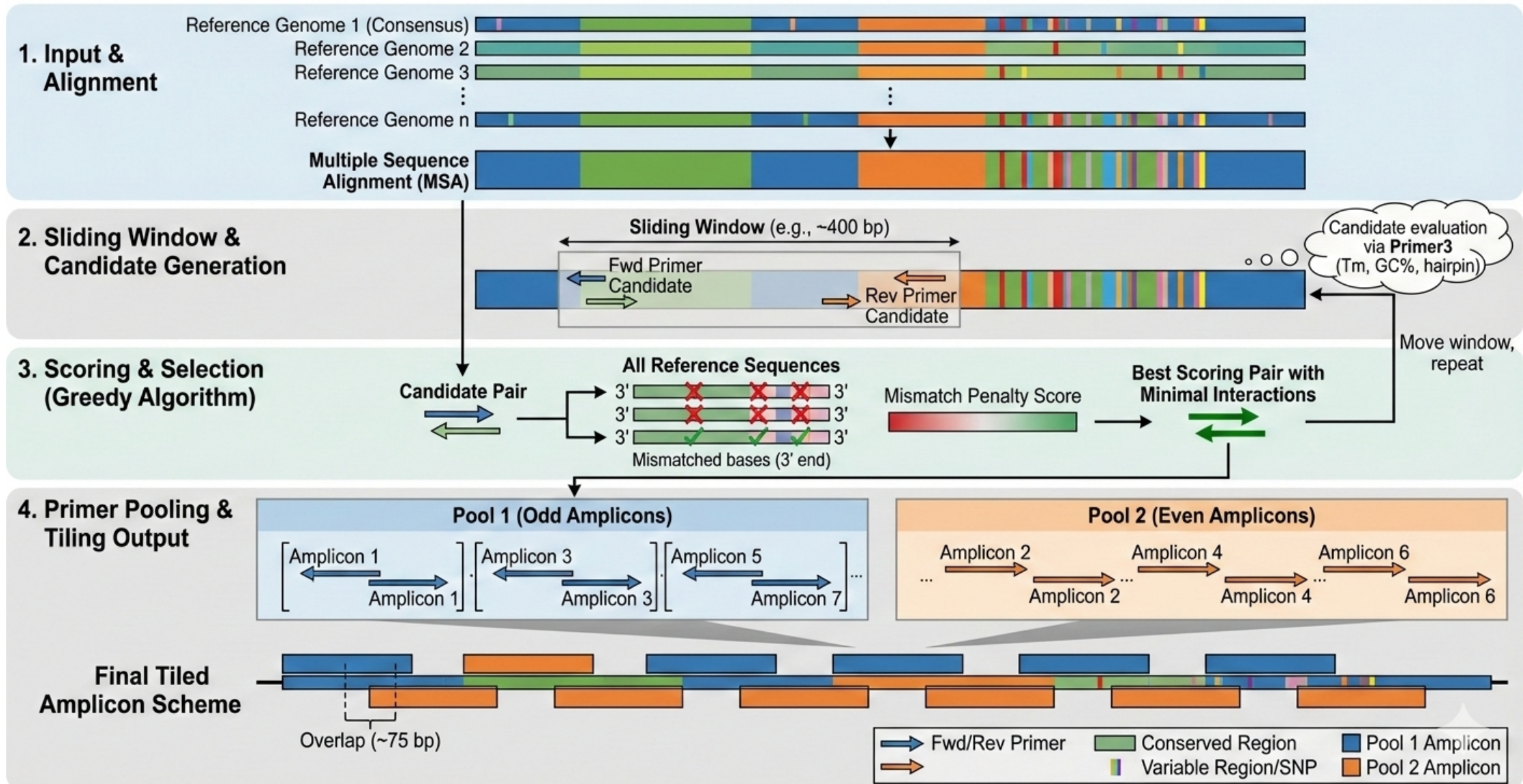
Juma, J.; Konongoi, S.L.; Nsengimana, I.; et al. Using Multiplex Amplicon PCR Technology to Efficiently and Timely Generate Rift Valley Fever Virus Sequence Data for Genomic Surveillance. *Viruses* (2023).

Tiled Amplicon Approach

Tiled amplicons Sequencing Workflow



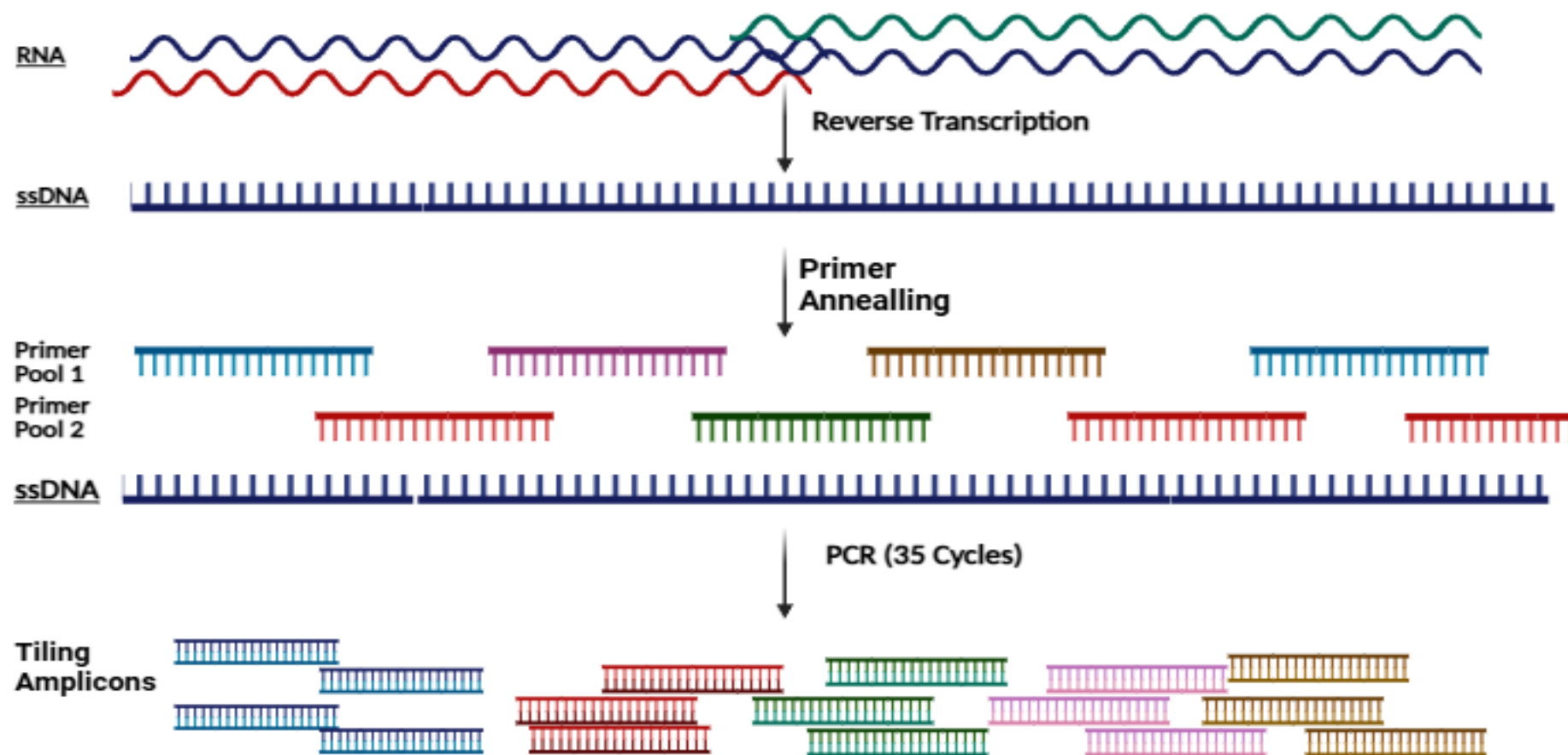
PrimalScheme: Amplicon Tiling Primer Design from Multiple Genomes



Web App: Primalcheme.com

Command Line [artic-network/primalscheme3](https://github.com/artic-network/primalscheme3)

Target Enrichment Step...



Advantages



High Sensitivity and Low Input Requirement.

It is highly effective for samples with low viral titers.



Direct Sequencing from Clinical Samples.

It eliminates the need for laborious laboratory viral culture.



Superior Signal-to-Noise Ratio.

The process selectively amplifies the viral genome while ignoring background host DNA or RNA.



Rapid Turnaround Time and Cost-effectiveness.

The protocol can be completed in 1–2 days and reduces the sequencing cost.



Comprehensive Coverage.

The primer design ensures that the entire coding sequence is recovered for downstream analysis.

Quality Checks & Screening

DenV-2 Virus tiled Amplicons

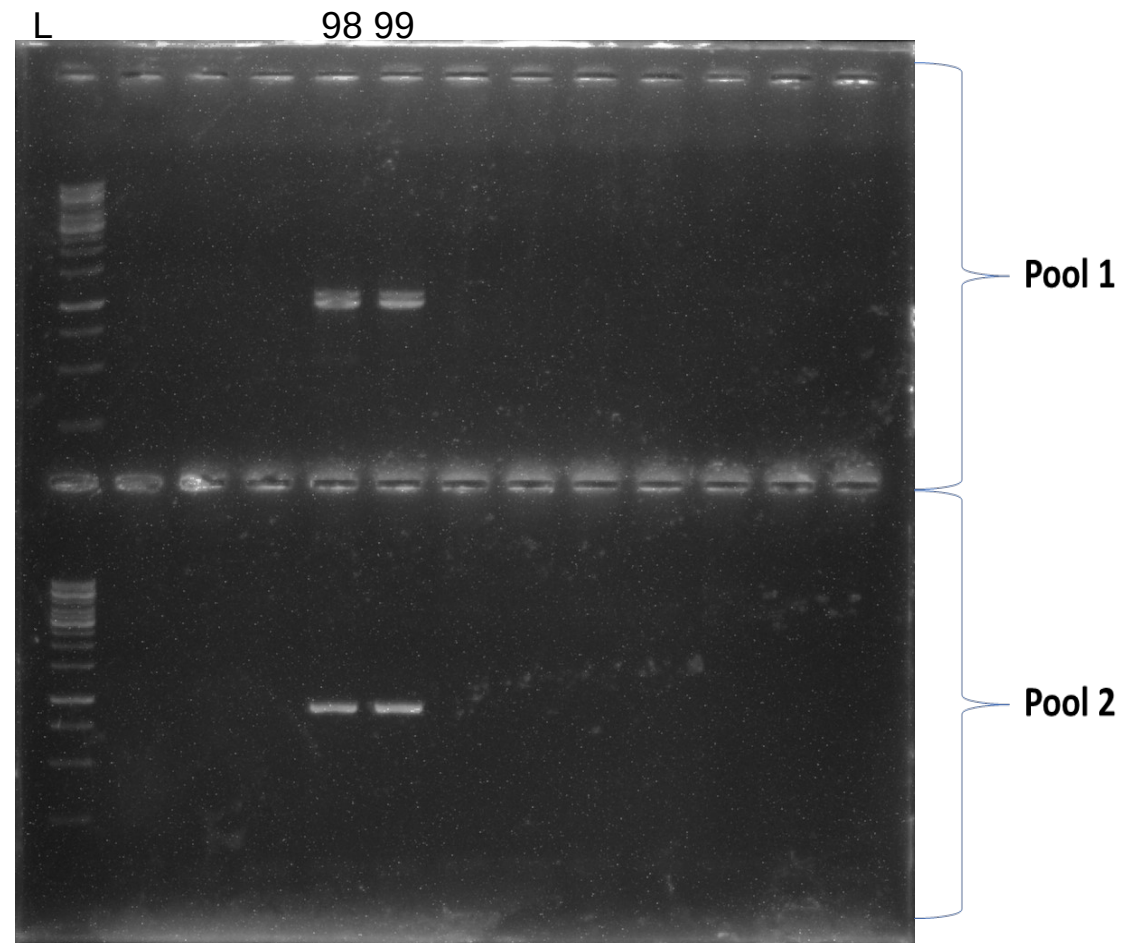
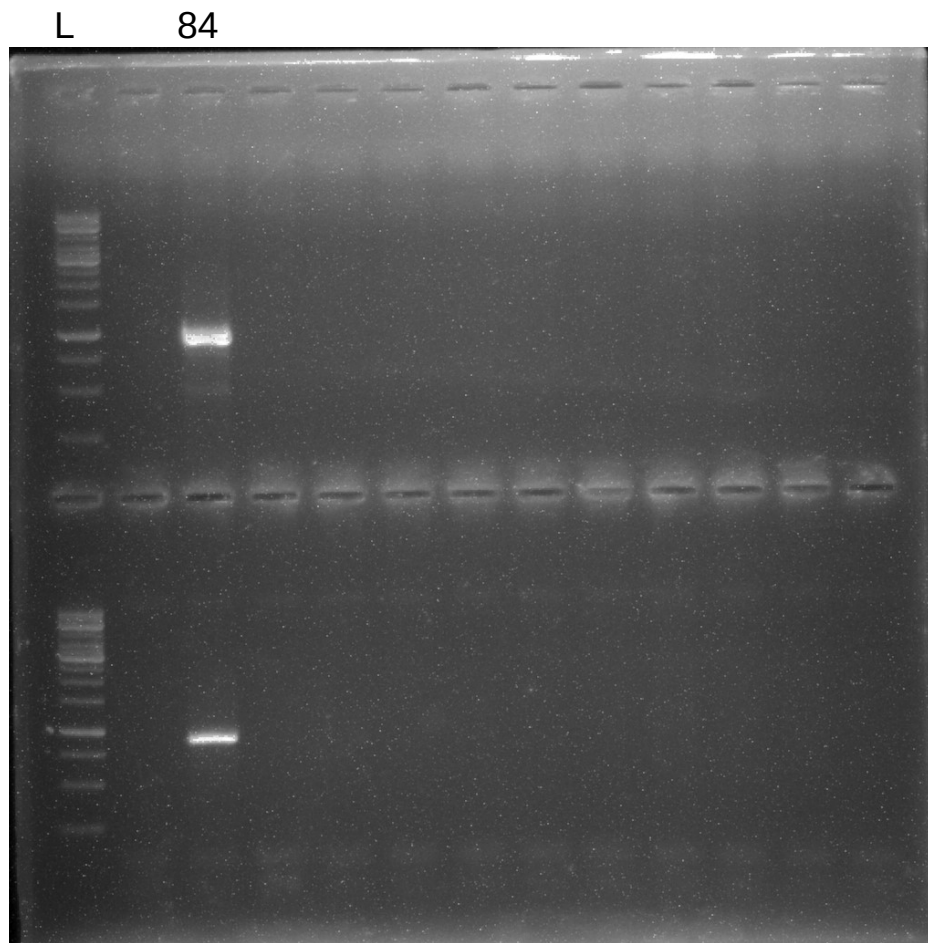
GeneRuler 1 kb DNA Ladder

bp ng/0.5µg %

10000	30	6
8000	30	6
6000	70	14
5000	30	6
4000	30	6
3500	30	6
3000	70	14
2500	25	5
2000	25	5
1500	25	5
1000	60	12
750	25	5
500	25	5
250	25	5

1% agarose

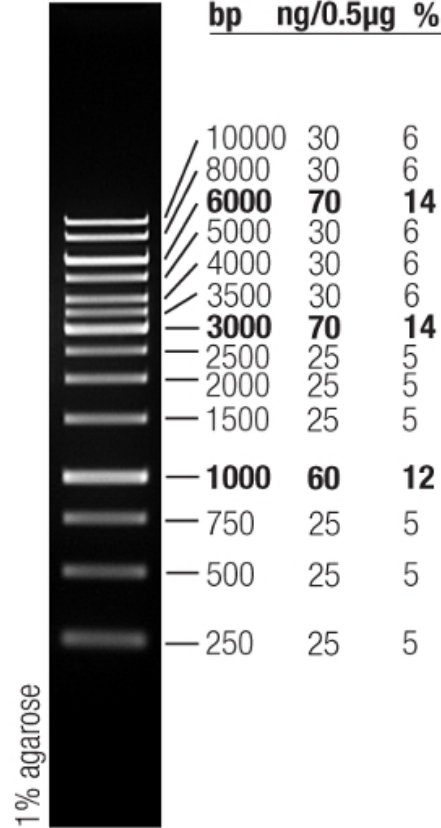
0.5 µg/lane, 8 cm length gel,
1X TAE, 7 V/cm, 45 min



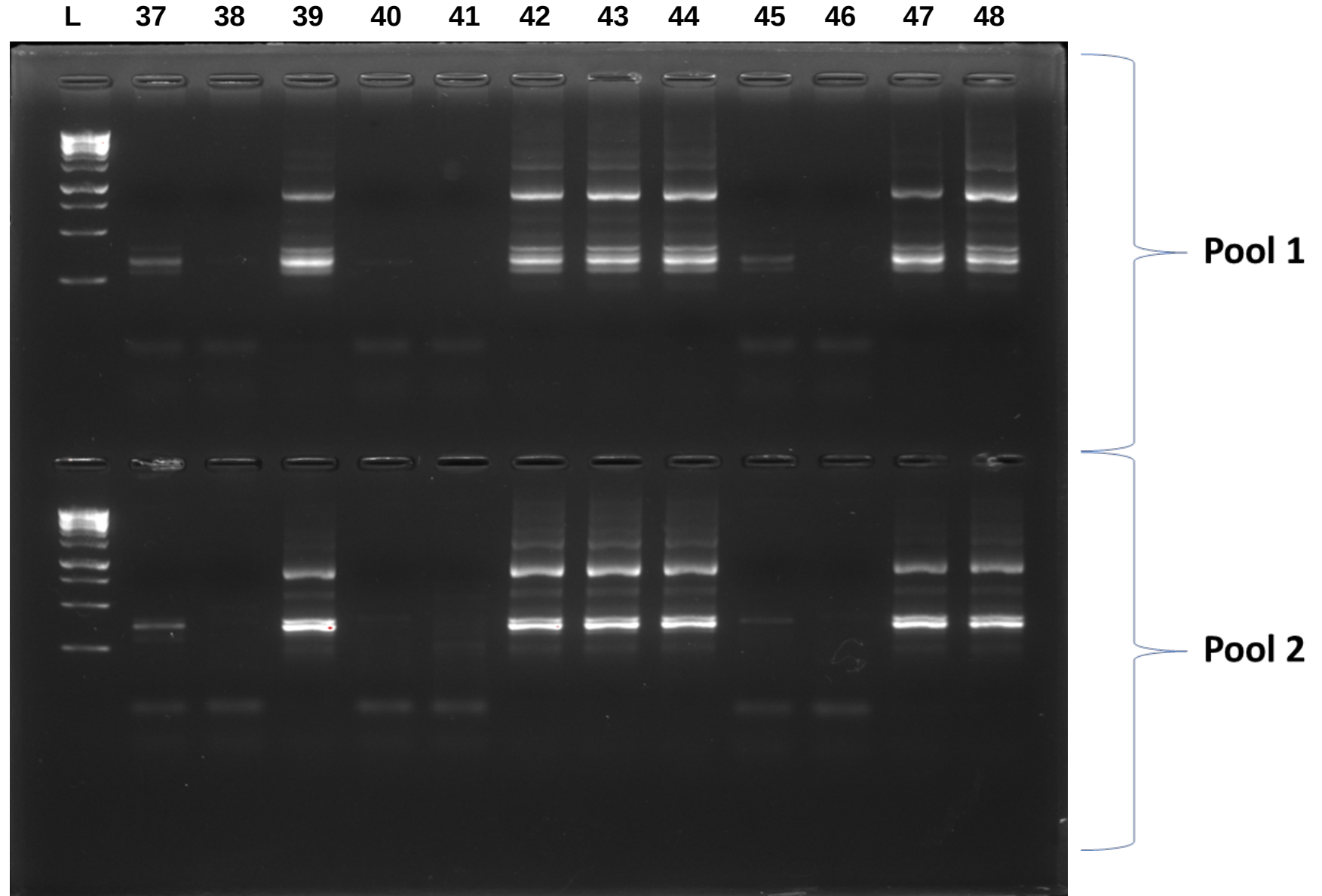
Chikungunya Virus tiled Amplicons

GeneRuler 1 kb DNA Ladder

bp ng/0.5µg %



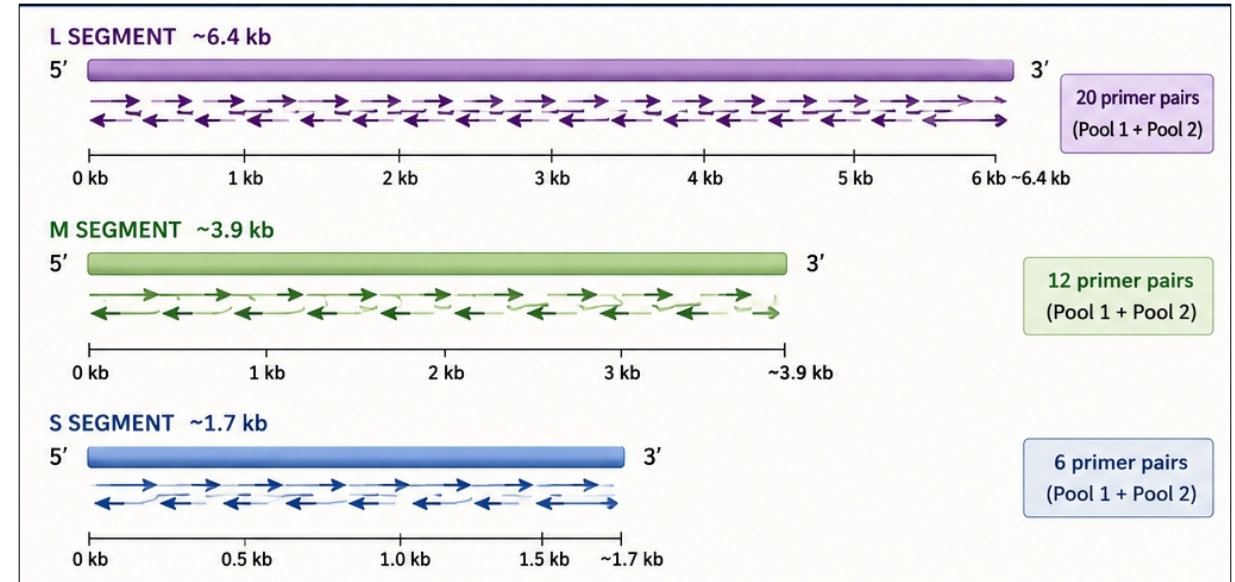
0.5 µg/lane, 8 cm length gel,
1X TAE, 7 V/cm, 45 min



Tiled Amplicon Approach: RVFV

RVFV Multiplex PCR Scheme

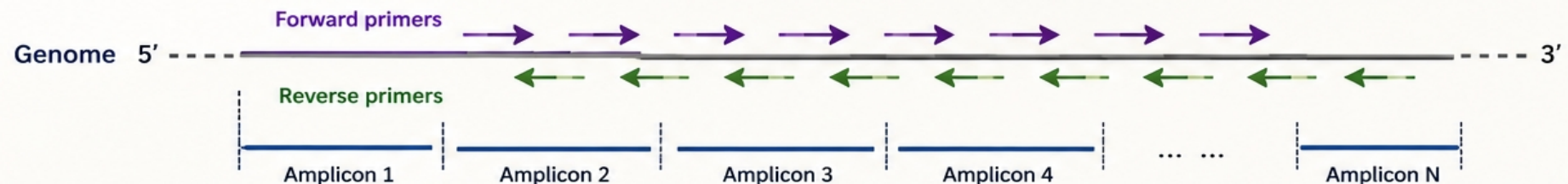
- **74 multiplex primers:** 38 (Pool 1) + 36 (Pool 2)
- Target amplicon size: ~400 bp (Optimal for short-read)
- Coverage by segment:
 - L segment: 20 primer pairs
 - M segment: 12 primer pairs
 - S segment: 6 primer pairs
- In silico predicted coverage: 97-99%



CONCEPT: Overlapping 'tiled' primers cover entire genome with **NO GAPS**

TILED PRIMER CONCEPT

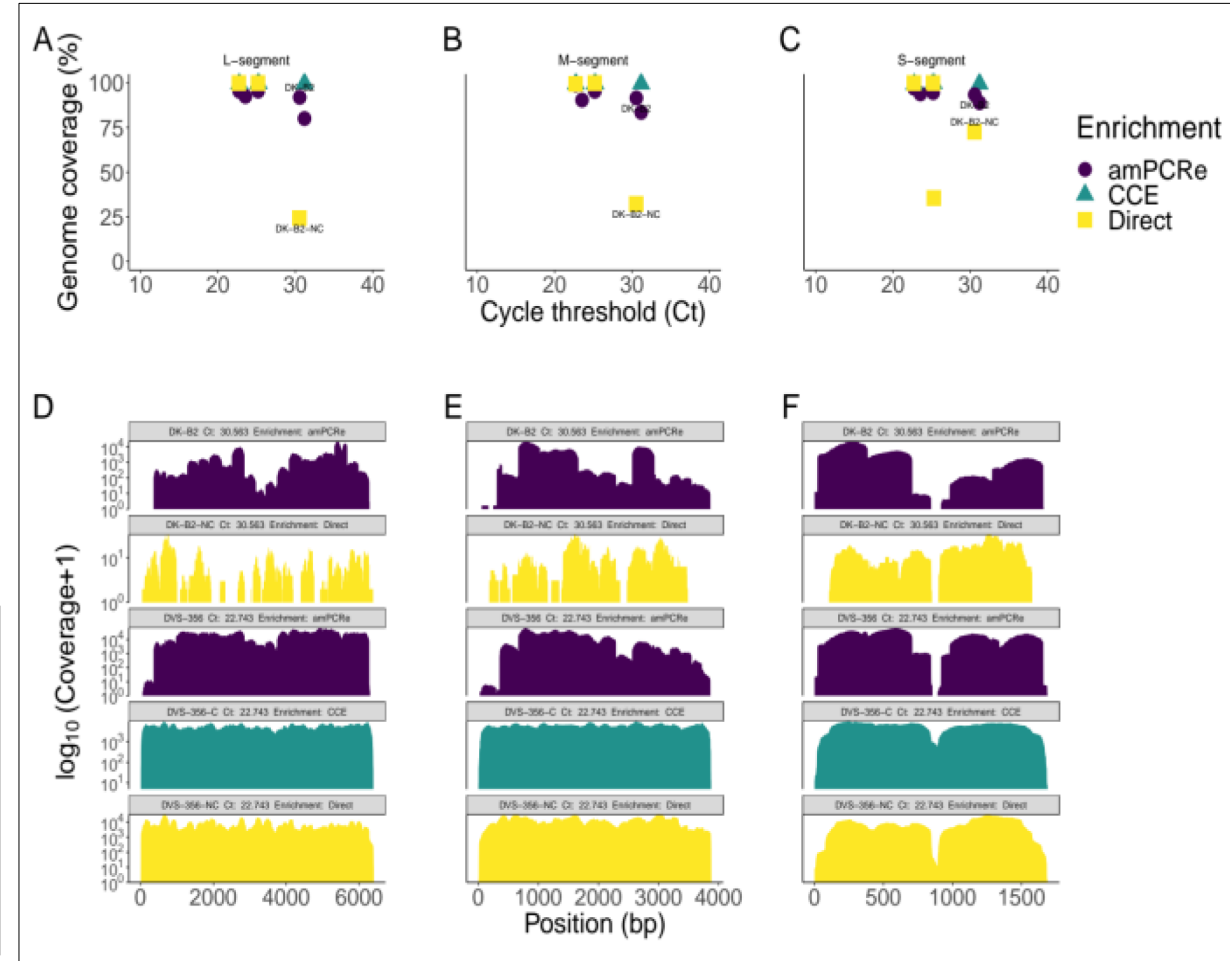
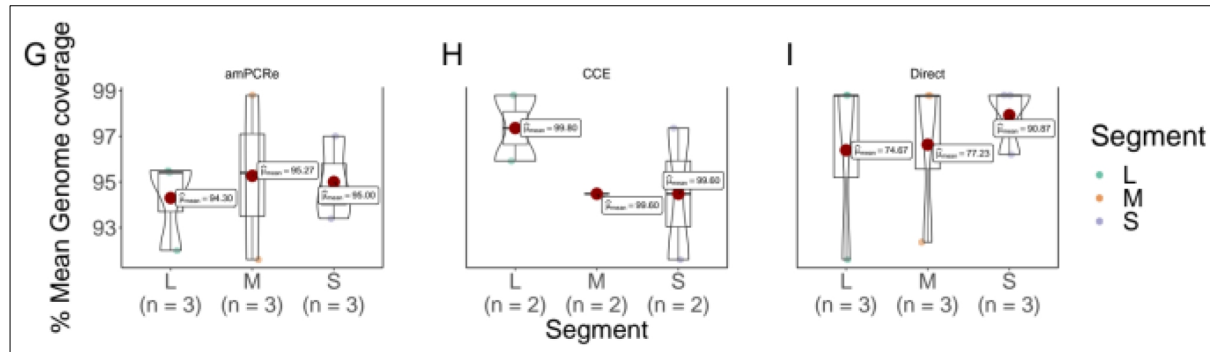
Overlapping amplicons ensure continuous coverage across the entire genome with **NO GAPS**



Amplicons (~400 bp) overlap to ensure continuous coverage with **NO GAPS**

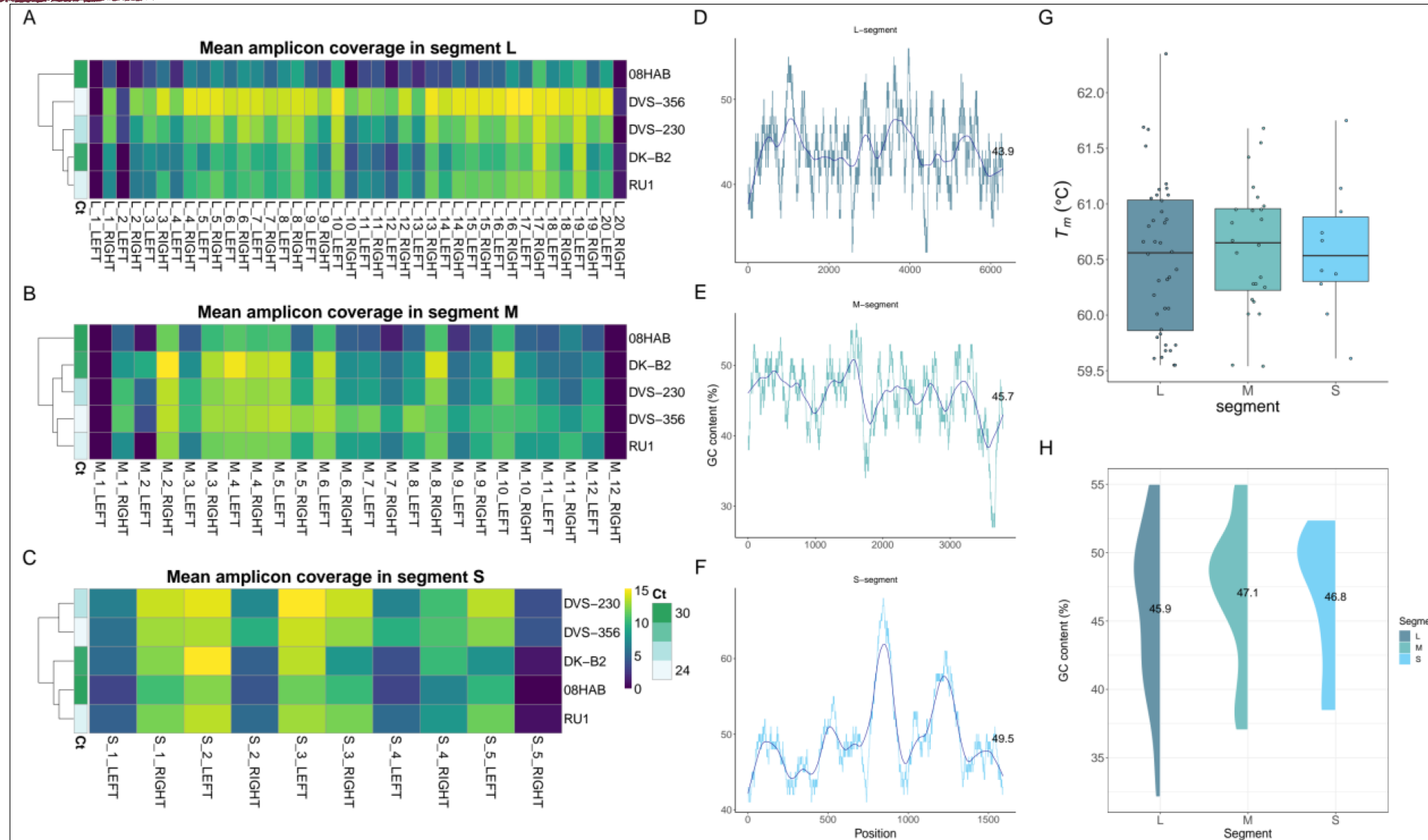
NGS Approaches: Genome Coverage Comparison

Study	Cattle samples - RVF outbreaks (n = 5)		
Approaches	CCE	amPCRe	Metagenomics
% Coverage	> 99%	80 - 97%	25 - 70 %
Ct Range	> 25	> 30	Failed for > 30
Time	7-14 days	24 - 48 hrs	****
~ Cost/Isolate	200 - 500	50 -100	****



Juma, J.; Konongoi, S.L.; Nsengimana, I.; et al. Using Multiplex Amplicon PCR Technology to Efficiently and Timely Generate Rift Valley Fever Virus Sequence Data for Genomic Surveillance. *Viruses* (2023).

Tiled Amplicon Approach: Genome Coverage



Juma, J.; Konongoi, S.L.; Nsengimana, I.; et al. Using Multiplex Amplicon PCR Technology to Efficiently and Timely Generate Rift Valley Fever Virus Sequence Data for Genomic Surveillance. *Viruses* (2023).

Conclusion

Tiled amplicon sequencing addresses key challenges:

- Enables detection on low viral titer
- Rapid, and
- cost-effective

NGS Whole-Genome Sequencing makes it possible for large-scale surveillance and outbreak response.

Bioinformatics Analysis: Lineage assignment and phylogenetic analysis

Why Phylogenetics (In Pathogen Genomics)?

In outbreak investigation:

- Determine source (local amplification vs. imported)
- Identify transmission chains
- Assess genetic relationship of outbreak isolates
- Detect zoonotic spillover events

In surveillance:

- Monitor temporal and spatial spread
- Detect introduction of new lineages
- Understand evolutionary dynamics
- Guide control strategy decisions

Phylogenetic Tree Construction



Methods used:

- Neighbor-Joining (NJ): Fast, good for large datasets
- Maximum Likelihood (ML): Statistically rigorous, assumes evolutionary model
- Bayesian: Probabilistic, includes uncertainty estimates



Software tools:

- IQ-TREE: Fast ML trees, bootstrap support
- MrBayes: Bayesian inference, posterior probabilities
- PAUP: Classic phylogenetics software



Reference sequences:

- 51 unique Gn gene sequences
- 47 unique whole genome sequences
- Chosen for bootstrap support >70%

Phylogenetic Tree Construction: RVFV

-  Web Application: <https://www.genomedetective.com/app/typingtool/rvf/>
-  Web Application: <https://www.genomedetective.com/db/ui/>
-  Command-line: <https://github.com/ajodeh-juma/rvftyping>

Raw Data:

1. GenBank SRA - [SRX24403625](https://hpc.ilri.cgiar.org/~gkibet/ndp_vip/) (https://hpc.ilri.cgiar.org/~gkibet/ndp_vip/)
2. Tutorial: **github - ndp-vip/RVFV Sequence Analysis Workflow**
3. GenBank SRA Search for Other RVFV sequences

Workflow:

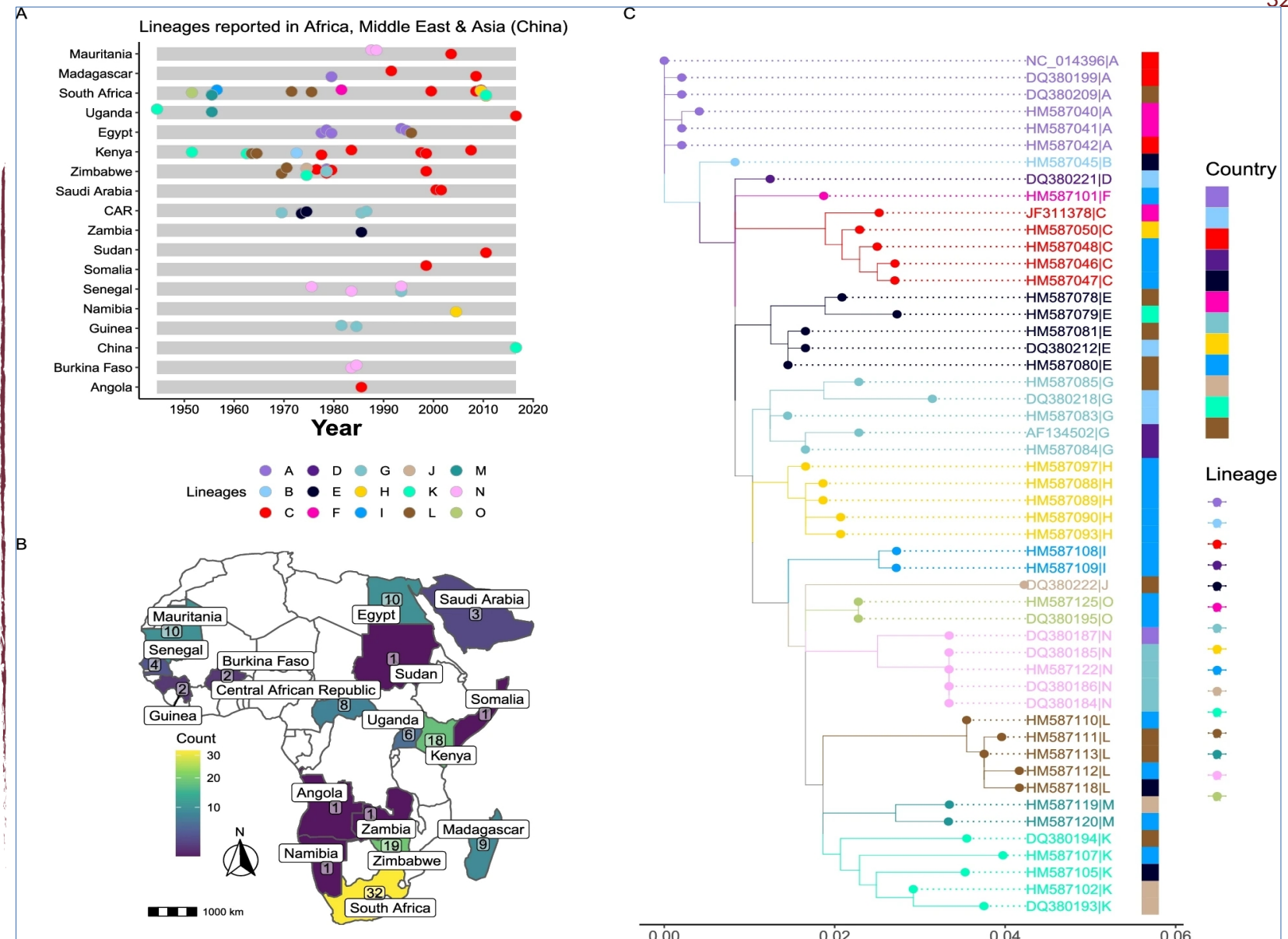
1. User submits sequence (Gn gene or whole genome)
2. DIAMOND search for species classification
3. MAFFT alignment to reference sequences
4. ML tree construction (IQ-TREE)
5. Lineage assignment if bootstrap >70%

Validation Results (Oyola et al. 2022):

- Gn gene: 100% accuracy on 129 sequences
- Whole genome: 99-100% accuracy on 234 sequences

What About RVFV?

- Lineages: 15 distinct lineages identified in Africa (A through O)
- Lineage C: Most predominant (documented since 1976)
- Lineage H: Second most common in recent outbreaks
- Other lineages: Geographically or temporally restricted



Juma, J., Fonseca, V., Konongoi, S.L. et al. Genomic surveillance of Rift Valley fever virus: from sequencing to lineage assignment. BMC Genomics 23, 520 (2022).

Acknowledgements

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- Paul Dobi
- Edward Kiritu - (Drafting/Wet-Lab)
- Kennedy Mwangi
- Shebbar Osiany



More about the lecture: <https://github.com/kibet-gilbert/rvf-v-ns-surveillance>

Tack så mycket!!!

Asante!!!

Email & Collaboration:

- Samuel O. Oyola (ILRI): S.Oyola@cgiar.org
- John A. Juma (ILRI): j.juma@cgiar.org
- Gilbert Kibet (Presentation Author)

Online Tools & Databases:

- RVFV Typing Tool: <https://www.genomedetective.com/app/typingtool/rvfv/>
- GitHub/Primal Scheme: https://github.com/artic-network/primal_schemes
- NCBI GenBank: <https://www.ncbi.nlm.nih.gov/genbank/>

Key Publications:

- Oyola et al. (2022) BMC Genomics - RVFV lineage assignment tool
- Juma et al. (2023) Viruses - Amplicon PCR methodology

Training Resources:

- Africa CDC Bioinformatic Training Portal
- Coursera NGS Analysis Courses
- EMBL-EBI Online Workshops